Apex Compute Unified Engine v1.1

Comprehensive Reference

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1 Introduction

This document provides a comprehensive reference for the AI compute operations supported by the engine. It details the specifications, formulas, and performance metrics for each operation. All benchmarks were collected on RTL running on a Kintex UltraScale+ FPGA in real time.

2 FPGA Presilicon Prototype Setup

2.1 System parameters

- Memory interface: DDR4 at 1333 MHz with a 32-bit data path.
- Engine frequency: 333 MHz.
- Memory interface clock: synchronized 1:1 with the engine clock.
- Data width: 256 bits.
- Total power consumption: 4.5 W.
- Total on-chip SRAM: 1.05 MBytes.

2.2 Peak operation rate

Total floating-point operations per second (FLOPS) from the engine at 333 MHz engine speed is approximately 42 GFLOPS/s.

2.3 FPGA resource utilization

Name	CLB LUTs	CLB Registers	Block RAM Tile	URAM	DSPs
unified_engine_top	78,348	50,045	16	30	197

3 FLOPS numbers

- FMA (Fused Multiply-Add): 2 FLOPS
- Addition/Multiplication: 1 FLOPS
- Exponent: 1 FLOPS
- Division: 1 FLOPS

4 Operation Table (BF16 domain)

BF16 operation benchmarks (including DMA in/out). Engine speed: 333 MHz; theoretical peak performance: 42 GFLOPS/s. Table metrics are based on $M = 1024$, $K = 1024$, $N = 1024$.

Table 1: BF16 operation benchmarks where $M = 1024$, $K = 1024$, $N = 1024$

Op	Operand dimensions	FLOPS (math)	Cycles (latency)	Achieved
AB^{T}	$A[M,K]$, $B[N,K] \to \mathrm{O}[M,N]$	$2MKN$	17,820,455 (53.3 ms)	40.17 GFLOPS/s
$AB^{\mathsf{T}}+C$	$A[M,K]$, $B[N,K]$, $C[M,N] \to \mathrm{O}[M,N]$	$2MKN + MN$	17,858,564 (53.5 ms)	40.10 GFLOPS/s
GELU(AB^{T})	$A[M,K]$, $B[N,K] \to \mathrm{O}[M,N]$	$2MKN + 4MN$	17,923,045 (53.7 ms)	40.02 GFLOPS/s
GELU($AB^{\mathsf{T}}+C$)	$A[M,K]$, $B[N,K]$, $C[M,N] \to \mathrm{O}[M,N]$	$2MKN + MN + 4MN$	17,927,850 (53.7 ms)	40.03 GFLOPS/s
SiLU(AB^{T})	$A[M,K]$, $B[N,K] \to \mathrm{O}[M,N]$	$2MKN + 4MN$	17,921,594 (53.7 ms)	40.02 GFLOPS/s
SiLU($AB^{\mathsf{T}}+C$)	$A[M,K]$, $B[N,K]$, $C[M,N] \to \mathrm{O}[M,N]$	$2MKN + MN + 4MN$	17,926,623 (53.7 ms)	40.03 GFLOPS/s
softmax(AB^{T})	$A[M,K]$, $B[N,K] \to \mathrm{O}[M,N]$	$2MKN + 5MN$	19,004,997 (57.01 ms)	37.76 GFLOPS/s
softmax($AB^{\mathsf{T}}+C$)	$A[M,K]$, $B[N,K]$, $C[M,N] \to \mathrm{O}[M,N]$	$2MKN + MN + 5MN$	19,051,310 (57.15 ms)	37.68 GFLOPS/s
A^{T}	$A[M,N] \to \mathrm{O}[N,M]$	0	1,648,647 (4.9 ms)	N/A
A·scalar	$A[M,N] \to \mathrm{O}[M,N]$	MN	180,500 (541 μs)	1.94 GFLOPS/s
A+scalar	$A[M,N] \to \mathrm{O}[M,N]$	MN	181,005 (543 μs)	1.93 GFLOPS/s
$A \cdot B$	$A[M,N]$, $B[M,N] \to \mathrm{O}[M,N]$	MN	263,580 (790 μs)	1.33 GFLOPS/s
$A+B$	$A[M,N]$, $B[M,N] \to \mathrm{O}[M,N]$	MN	263,871 (791 μs)	1.33 GFLOPS/s
RMSNorm(A)·γ	$A[M,N]$, $\gamma[N] \to \mathrm{O}[M,N]$	$4MN$	290,945 (872 μs)	4.81 GFLOPS/s
LayerNorm(A)·$\gamma+\beta$	$A[M,N]$, $\gamma[N]$, $\beta[N] \to \mathrm{O}[M,N]$	$7MN$	414,679 (1.24 ms)	5.90 GFLOPS/s

Note: O denotes the output tensor.

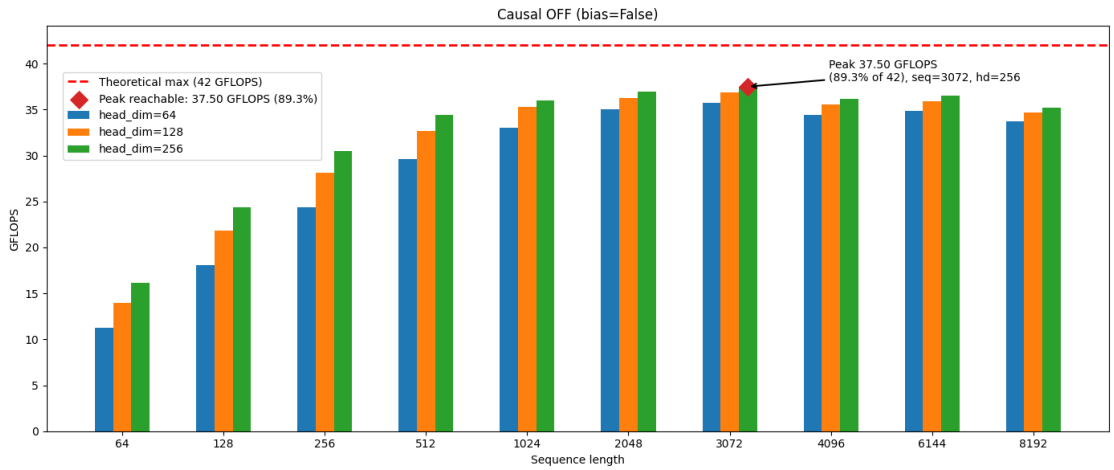

Figure 1: Bias off

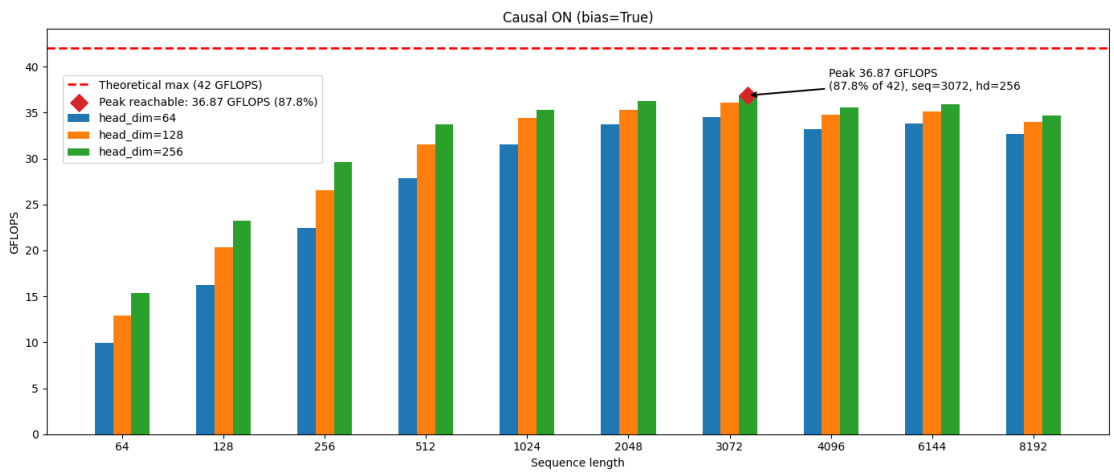

Figure 2: Bias on

Memory-efficient attention FLOPs: The following kernel is written for the engine to compute the attention block for given query/key/value tensors and an optional mask or bias. It reaches almost 90% utilization of theoretical FLOPS.

$$\texttt{memory_efficient_attention}(q, k, v, \texttt{mask_or_bias}).$$

The kernel is equivalent to the following PyTorch reference implementation.

```
def memory_efficient_attention(q, k, v, attn_bias=None):
    scale = 1.0 / math.sqrt(head_dim)
    attn_weights = (q @ k.T) * scale
    if attn_bias is not None:
        attn_weights = attn_weights + attn_bias
    scores = torch.softmax(attn_weights, dim=-1)
    return scores @ v
```

5 Operation Table (Quantized domain)

Quantized operation benchmarks (including DMA in/out). Engine speed: 333 MHz; theoretical peak: 42 GFLOPS/s. In quantized mode, achieved FLOPS are the same **for any M**. In contrast, for tiled matrix-matrix multiplication, smaller M reduces FLOPS utilization. fp4 refers to nvfp4(Nvidia fp4).

Table 2: Quantized-domain operation list where $M = 1024$, $K = 1024$, $N = 1024$.

Op	Precision	Operand dimensions	FLOPS (math)	Cycles (latency)	Achieved
AB^{T}	A(bf16) B(int4/fp4) O(bf16)	$A[M,K]$, $B[N,K] \to \mathrm{O}[M,N]$	$2MKN$	22,849,177 (68.5 ms)	31.33 GFLOPS/s
$AB^{\mathsf{T}}+C$	A(bf16) B(int4/fp4) C(bf16) O(bf16)	$A[M,K]$, $B[N,K]$, $C[M,N] \to \mathrm{O}[M,N]$	$2MKN + MN$	23,073,635 (69.2 ms)	31.04 GFLOPS/s
GELU(AB^{T})	A(bf16) B(int4/fp4) O(bf16)	$A[M,K]$, $B[N,K] \to \mathrm{O}[M,N]$	$2MKN + 4MN$	22,850,336 (68.5 ms)	31.39 GFLOPS/s
GELU($AB^{\mathsf{T}}+C$)	A(bf16) B(int4/fp4) C(bf16) O(bf16)	$A[M,K]$, $B[N,K]$, $C[M,N] \to \mathrm{O}[M,N]$	$2MKN + MN + 4MN$	23,100,231 (69.3 ms)	31.06 GFLOPS/s
SiLU(AB^{T})	A(bf16) B(int4/fp4) O(bf16)	$A[M,K]$, $B[N,K] \to \mathrm{O}[M,N]$	$2MKN + 4MN$	22,850,243 (68.5 ms)	31.39 GFLOPS/s
SiLU($AB^{\mathsf{T}}+C$)	A(bf16) B(int4/fp4) C(bf16) O(bf16)	$A[M,K]$, $B[N,K]$, $C[M,N] \to \mathrm{O}[M,N]$	$2MKN + MN + 4MN$	23,104,094 (69.3 ms)	31.06 GFLOPS/s

Table 3: Quantization/Dequantization ops where $N = 131072$.

Op	Precision	Operand dimensions	FLOPS (math)	Cycles (latency)	Achieved
Quantize(A)	A(bf16) O(int4/fp4)	$A[N] \to \mathrm{O}[N]$	$2N$	15266 (45.8 μs)	5.72 GFLOPS/s
Dequantize(A)	A(int4/fp4) O(bf16)	$A[N] \to \mathrm{O}[N]$	N	13193 (39.5 μs)	3.31 GFLOPS/s